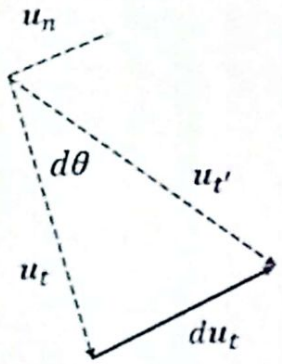
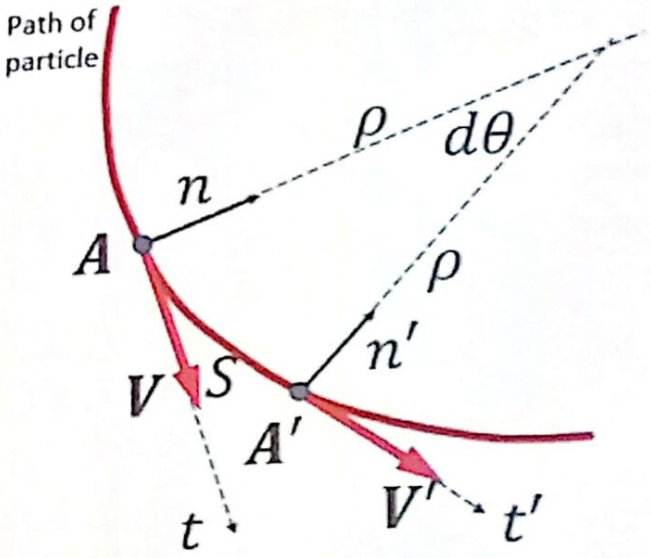
$$\frac{du_t}{dt} = \frac{d\theta}{dt}u_n$$
$$= \dot{\theta}u_n$$
$$= \frac{v}{\rho}u_n$$





Acceleration:

$$a = \frac{dV}{dt} = \frac{d(vu_t)}{dt}$$

$$= \frac{dv}{dt}u_t + v\frac{du_t}{dt}$$

a_t ⇑

$$\ldots = \dot{v}u_t + v\,\dot{\theta}u_n \quad = \quad \dot{v}u_t + \frac{v^2}{\rho}u_n$$

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$$a = \dot{v}u_t + \frac{v^2}{\rho}u_n$$

Thus,

$$a = a_t u_t + a_n u_n$$

Where, $a_t = \dot{v}$ Tangential component

$$a_t = v\frac{dv}{ds}$$

$$a_n = \frac{v^2}{\rho}$$ Normal component

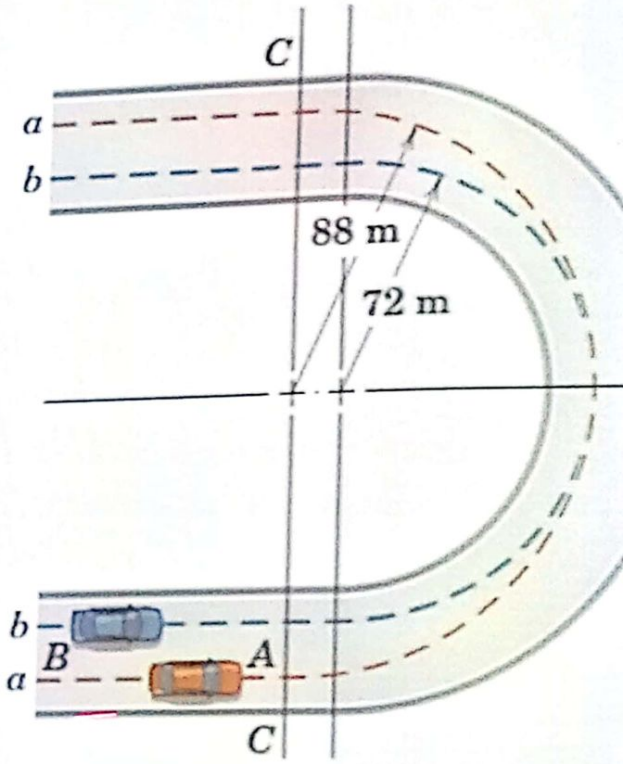
With magnitude of

$$a = \sqrt{(a_t)^2 + (a_n)^2}$$

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Example

Race car A follows path a-a while race car B follows path b-b on the unbanked track. If each car has a constant speed limited to that corresponding to a lateral (normal) acceleration of $0.8g$, determine the times t_A and t_B for both cars to negotiate the turn as delimited by the line C-C.

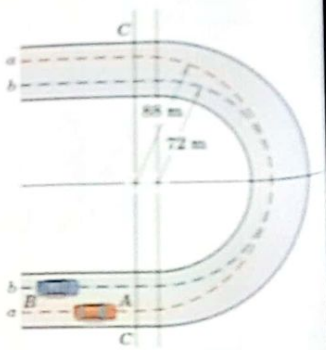


Solution

$$a_n = 0.8\,g = \frac{v^2}{\rho} \quad \rightarrow \quad v = \sqrt{0.8\,g\,\rho}$$

Car A: $v_A = \sqrt{0.8\,(9.81)(88)} = 26.3 \quad m/s$

Car B: $v_B = \sqrt{0.8\,(9.81)(72)} = 23.8 \quad m/s$



For constant speed:

$$t_A = \frac{S_A}{v_A} = \frac{\pi(88)}{26.3} = 10.52$$

$$t_B = \frac{S_B}{v_B} = \frac{\pi(72) + 2(16)}{23.8} = 10.86 \quad s$$

Car A will win the race !

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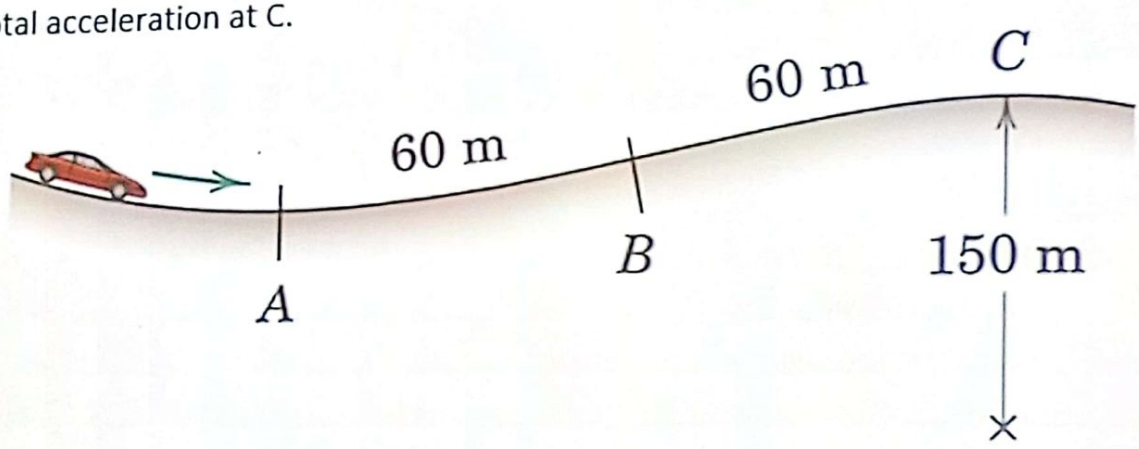
Example

To anticipate the dip and hump in the road, the driver of a car applies brakes to produce a uniform deceleration. The speed is 100 km/h at the bottom A of the dip and 50 km/h at the top C of the hump, which is 120 m along the road from A. If the passengers experience a total acceleration of 3 m/s² at A and if the radius of curvature of the hump at C is 150 m, calculate

(a) the radius of curvature at A,

(b) the acceleration at the inflection point B, and

(c) the total acceleration at C.



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Given:

$$v_A = 100\left(\frac{1000}{3600}\right) = 27.8\ m/s \qquad S_{AC} = 120\ m$$

$$v_C = 50\left(\frac{1000}{3600}\right) = 13.89\ m/s \qquad a_A = 3\ m/s^2 \text{ (total)}$$

$$\rho_C = 150\ m$$

The acceleration is uniform along the path (constant):

$$v^2 = v_0^2 + 2aS \longrightarrow (v_c)^2 = (v_A)^2 + 2a_t\, S_{AC}$$

$$\therefore a_t = \frac{1}{2S_{AC}}\left((v_c)^2 - (v_A)^2\right)$$

$$\therefore a_t = \frac{1}{2(120)}\left((13.89)^2 - (27.8)^2\right)$$

$$\therefore a_t = -2.41\ \ m/s^2$$

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(a) Condition at A:

since: $a_A = 3\,\frac{m}{s^2}$ (given)

and: $a_t = -2.41\,\frac{m}{s^2}$ (calculated)

$$a = \sqrt{(a_n)^2 + (a_t)^2} \quad \rightarrow \quad a_n = \sqrt{(a_A)^2 - (a_t)^2}$$

$$= \sqrt{(3)^2 - (-2.41)^2} \quad = 1.785\,\frac{m}{s^2}$$

$$a_n = \frac{v^2}{\rho} \quad \rightarrow \quad \rho = \frac{v^2}{a_n} \quad = \frac{(27.8)^2}{1.785} = 432 \;\; m$$

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(b) Condition at B.

The radius of curvature is infinite at the inflection

$$a_n = \frac{v^2}{\infty} = 0$$

$$\therefore a = a_t = -2.41 \ \ m/s^2$$

(c) Condition at C: The normal acceleration becomes

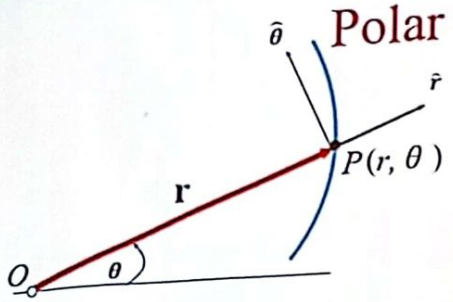
$$a_n = \frac{v^2}{\rho} \rightarrow a_n = \frac{(13.89)^2}{150} = 1.286 \ \ \text{m/s}^2$$

$$a = \sqrt{(a_n)^2 + (a_t)^2} \rightarrow a = \sqrt{(1.286)^2 + (-2.41)^2}$$

$$= 2.73 \ \frac{m}{s^2}$$

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Polar Coordinates



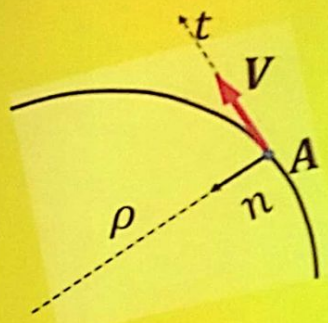
The Position $\boldsymbol{r = r\hat{r}}$

The Velocity $\mathbf{V} = \dfrac{d\boldsymbol{r}}{dt}$

$$\boldsymbol{V = v_r\hat{r} + v_\theta\hat{\theta}}$$

$$v_r = \dot{r}$$

$$v_\theta = r\dot{\theta}$$



$$v = \rho\,\dot{\theta}$$

$$\mathbf{V} = v\boldsymbol{u}_t$$

$$\mathbf{a} = \dot{v}\boldsymbol{u}_t + \frac{v^2}{\rho}\boldsymbol{u}_n$$

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Acceleration

$$\boldsymbol{a} = a_r\,\hat{\boldsymbol{r}} + a_\theta\,\hat{\boldsymbol{\theta}}$$

$$a_r = \ddot{r} - r\dot{\theta}^2$$

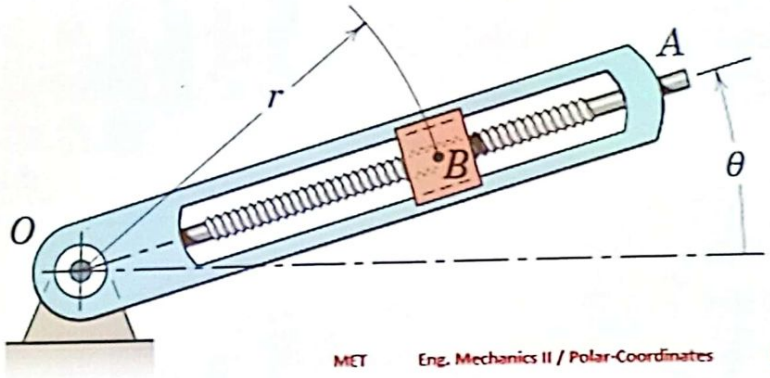
$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

$$a = \sqrt{a_r^2 + a_\theta^2}$$

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Example

Rotation of the slotted arm is governed by $\theta = 0.2\,t + 0.02t^3$, where θ is in radians and t is in seconds. Simultaneously, distance of the slider B from O is controlled according to $r = 0.2 + 0.04t^2$, where r is in meters and t is in seconds. Calculate the magnitudes of the velocity and acceleration of the slider for the instant when $t = 3$ s.



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Radial Direction

$$r = 0.2 + 0.04t^2 \longrightarrow r|_{t=3} = 0.2 + 0.04(3)^2 = 0.56\ m$$

$$\dot{r} = \frac{dr}{dt} = 0.08t \longrightarrow \dot{r}\Big|_{t=3} = 0.08(3) = 0.24\ m/s$$

$$\ddot{r} = \frac{d\dot{r}}{dt} = 0.08$$

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Radial Direction

$$r = 0.2 + 0.04t^2 \longrightarrow r|_{t=3} = 0.2 + 0.04(3)^2 = 0.56\ m$$

$$\dot{r} = \frac{dr}{dt} = 0.08t \longrightarrow \dot{r}\Big|_{t=3} = 0.08(3) = 0.24\ m/s$$

$$\ddot{r} = \frac{d\dot{r}}{dt} = 0.08 \longrightarrow \ddot{r}\Big|_{t=3} = 0.08\ m/s^2$$

Transverse Direction

$$\theta = 0.2t + 0.02t^3 \longrightarrow \theta|_{t=3} = 0.2(3) + 0.02(3)^3 = 1.14\ rad$$

$$\dot{\theta} = \frac{d\theta}{dt} = 0.2 + 0.06t^2 \longrightarrow \dot{\theta}\Big|_{t=3} = 0.2 + 0.06(3)^2 = 0.74\ rad/s$$

$$\ddot{\theta} = \frac{d\dot{\theta}}{dt} = 0.12t \longrightarrow \ddot{\theta}\Big|_{t=3} = 0.12(3) = 0.36\ rad/s^2$$

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$r = 0.56\ m$ $\quad \theta = 1.14\ rad$

$\dot{r} = 0.24\ m/s$ $\quad \dot{\theta} = 0.74\ rad/s$

$\ddot{r} = 0.08\ m/s^2$ $\quad \ddot{\theta} = 0.36\ rad/s^2$

The velocity components:

$$v_r = \dot{r}; \quad \Rightarrow \quad v_r = 0.24 \quad m/s$$

$$v_\theta = r\dot{\theta}; \quad \Rightarrow \quad v_\theta = 0.56(0.74) = 0.414 \quad m/s$$

Thus,

$$v = \sqrt{(v_r)^2 + (v_\theta)^2} \Rightarrow v = \sqrt{(0.24)^2 + (0.414)^2} = 0.479 \quad m/s$$

The acceleration components:

$$a_r = \ddot{r} - r\dot{\theta}^2; \quad \Rightarrow \quad a_r = 0.08 - 0.56(0.74)^2 = -0.227\ m/s^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}; \quad \Rightarrow \quad a_\theta = 0.56(0.36) + 2(0.24)(0.74) = 0.557 \quad m/s^2$$

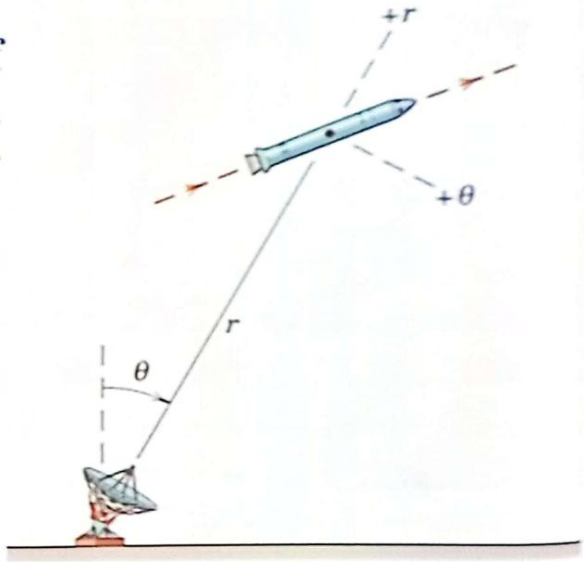
Thus,

$$a = \sqrt{(a_r)^2 + (a_\theta)^2} \Rightarrow a = \sqrt{(-0.227)^2 + (0.557)^2} = 0.601 \quad m/s^2$$

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Example

A tracking radar lies in the vertical plane of the path of a rocket which is coasting in unpowered flight above the atmosphere. For the instant when $\theta = 30^0$, the tracking data give $r = 25(10^4)$ ft, $\dot{r} = 4000$ ft/sec, and $\dot{\theta} = 0.80$ deg/sec. The acceleration of the rocket is due only to gravitational attraction and for its particular altitude is 31.4 ft/sec^2 vertically down. For these conditions determine the velocity v of the rocket and the values of $\ddot{r}$ and $\ddot{\theta}$



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Solution

The velocity components:

$$v_r = \dot{r}; \quad \Rightarrow \quad v_r = 4000 \ \ ft/s$$

$$v_\theta = r\dot{\theta}; \quad \Rightarrow \quad v_\theta = 25(10^4)(0.8)\left(\frac{\pi}{180}\right)$$

$$= 3490 \ \ ft/s$$

Thus,

$$v = \sqrt{(v_r)^2 + (v_\theta)^2};$$

$$\Rightarrow \quad v = \sqrt{(4000)^2 + (3490)^2}\ ft/s$$

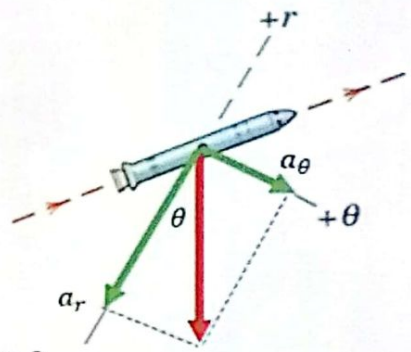
$$= 5310 \ \ ft/s$$

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Since the total acceleration of the rocket is $g = 31.4$ ft/sec^2 down ward, we resolve g in the r- and θ −directions

$$a_r = -31.4\cos 30 = -27.2\ ft/s^2$$

$$a_\theta = 31.4\sin 30 = 15.7\ ft/s^2$$



Now,

$$a_r = \ddot{r} - r\dot{\theta}^2; \implies -27.2 = \ddot{r} - 25(10^4)\left((0.8)\left(\frac{\pi}{180}\right)\right)^2$$

$$\therefore \ddot{r} = -27.2 + 25(10^4)\left((0.8)\left(\frac{\pi}{180}\right)\right)^2 = 21.5\ ft/s^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}; \implies 15.7 = 25(10^4)\ddot{\theta} + 2(4000)(0.8)\left(\frac{\pi}{180}\right)$$

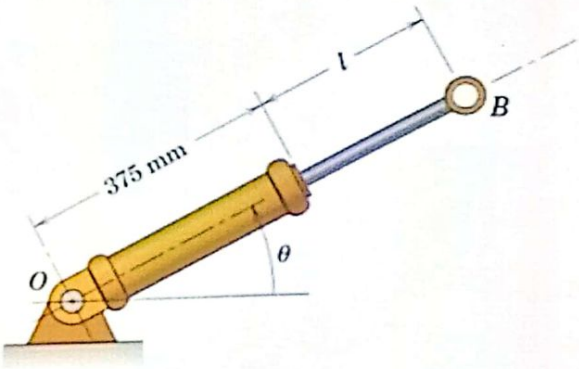
$$\therefore \ddot{\theta} = \frac{1}{25(10^4)}\left[15.7 - 2(4000)(0.8)\left(\frac{\pi}{180}\right)\right]$$

$$= -3.84(10^{-4})\ rad/s^2$$

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Example

If the cylinder rotates at the constant rate $\dot{\theta} = 60\ deg/s$ and l *is decreasing at the constant rate of 150* mm/s, calculate the magnitudes of the velocity **v** and acceleration **a** of end B ***when*** $l=125\ mm$



$\dot{\theta} = 60\left(\frac{\pi}{180}\right)$ rad/s

$\ddot{\theta} = 0$

$r = 375 + 125 = 500\ mm$

$\dot{r} = -150\ mm/s$

$\ddot{r} = 0$

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Solution

$\dot{\theta} = 60\left(\frac{\pi}{180}\right)$ rad/s

$\ddot{\theta} = 0$

$r = 375 + 1\dot{2}5 = 500\ mm$

$\dot{r} = -150\ mm/s$

$\ddot{r} = 0$

$v_r = \dot{r};$ ⟹ $v_r = -150\ \ mm/s$

$v_\theta = r\dot{\theta};$ ⟹ $v_\theta = 500(60)\left(\frac{\pi}{180}\right) = 523.3\ mm/s$

$v = \sqrt{(v_r)^2 + (v_\theta)^2};$ $v = \sqrt{(-150)^2 + (523.3)^2}$

$v = 544.4\ \text{mm/s}$

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$\dot{\theta} = 60\left(\frac{\pi}{180}\right)$ rad/s

$\ddot{\theta} = 0$

$r = 375 + 125 = 500\ mm$

$\dot{r} = -150\ mm/s$

$\ddot{r} = 0$

$a_r = \ddot{r} - r\dot{\theta}^2;$ ⟹ $a_r = 0 - 500\left(60\left(\frac{\pi}{180}\right)\right)^2;$

$= 0 - 500(1.047)^2$

$= -547.76\ mm/s^2;$

$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta};$ ⟹ $a_\theta = 0 + 2(-150)(1.047);$

$a_\theta = -314.1 \quad mm/s^2;$

$a = \sqrt{(a_r)^2 + (a_\theta)^2};$ ⟹ $a = \sqrt{(-547.76)^2 + (-314.1)^2}$

$= 631.43\ mm/s^2$

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Circular Motion

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